



KAZAKH-BRITISH
TECHNICAL
UNIVERSITY

APPROVED BY

Dean of SITE

Azamat Imanbayev

2025



SYLLABUS

Academic Year 2025-2026

Semester Fall

Discipline: INFT1101 Information communication technology

Volume of credits: 3 credits, 5 ECTS,

Course/Syllabus designer : Nursultan Yergazizuly Alpysbay

Personal Information about the Instructor	Time and place of classes		Contact information
	Lessons	Office Hours	e-mail
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COURSE DURATION: 15 weeks, 15 class hours

1. Course Overview

This course is designed to provide students with practical knowledge of computer concepts and basic skills necessary for work and communication in modern society. Students will study all the necessary Microsoft Office programs, databases, Power BI and the Python programming language.

2. Aims and objectives

This course is designed for absolute beginners in programming. It is expected that students will be able to work with tables, create high-quality presentations, edit the necessary file from the database, download and install software, as well as understand the basic concepts of programming.

3. Learning outcomes

After successful completion of this course, students will be able to do something useful with the help of information and communication technologies (ICT). Here are some of the practical skills and actions they will be able to perform:

- Data Analysis:
- Data Visualization:
- Data Integration:
- Process Automation:
- Python Programming:
- Preparation and Presentation:

In general, this course will provide students with the skills and knowledge necessary to work with data and information, which can be applied in various fields, including business, science, education and many others.

4. Textbooks and readings

Primary textbook

1. <https://uni-x.almv.kz/platform/login>

Supplementary textbooks

1. Lutz M. Learning python: Powerful object-oriented programming. – " O'Reilly Media, Inc.", 2013.
2. Chun W. Core python programming. – Prentice Hall Professional, 2001. – T. 1.
3. <https://www.w3schools.com/python/default.asp>
4. <https://www.coursera.org/programs/media-communications-and-history-of-kazakhstan-jur-bachelors-3jowb/learn/microsoft-excel-work-smarter?collectionId=skill~microsoft-excel>
5. <https://www.coursera.org/programs/media-communications-and-history-of-kazakhstan-jur-bachelors-3jowb/learn/microsoft-powerpoint-work-smarter>
6. <https://powerbi.microsoft.com/ru-ru/>
7. <https://www.postgresql.org>
8. https://edurev.in/courses/14282_Postgresql-A-Complete-Tutorial

5. Lesson Program

Week	Classes				
	Topic	Lecture	Laboratory	Tutorial	Textbook Chapter
1	Lecture #1. Introduction to ICT	2	1		Lecture #1
2	Lecture #2. Microsoft Excel Basics	2	1	Lab 1	Lecture #2
3	Lecture #3. Advanced functions and macros in Excel	2	1		Lecture #3
4	Lecture #4. Microsoft Power Point Basics	2	1	Lab 2	Lecture #4
5	Lecture #5. Interactive presentations with Microsoft Power Point	2	1	Quiz 1	Lecture #5
6	Lecture #6. SQL Basics	2	1	Lab 3	Lecture #6
7	Lecture #7. Working with Data in SQL	2	1		Lecture #7
8	Lecture #8. Basics of Python programming language	2	1	Midterm exam	Lecture #8
9	Lecture #9. Variables, types, and conditional operator	2	1	Lab 4	Lecture #9
10	Lecture #10. Loop in Python	2	1		Lecture #10
11	Lecture #11. List and Two-Dimensional Lists	2	1	Lab 5	Lecture #11
12	Lecture #12. Functions	2	1	Quiz 2	Lecture #12
13	Lecture #13. Introduction to Power BI	2	1	Lab 6	Lecture #13
14	Lecture #14. Data Visualization in Power BI	2	1		Lecture #14
15	Lecture #15. Integration SQL and Power BI	2	1	Endterm exam	Lecture #15

6. Course Requirements and Grades

COURSE ASSESSMENT PARAMETERS

Attendance	0 %
Laboratory works	24 %
Quizzes	10 %
Midterm/Endterm	26 %

Final exam	
Total	40 %
	100 %

Course assessment schedule

№	Assessment criteria	Weeks																Total
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 17	
1.	Attendance has to be more than 70%																	
2.	Activity on seminars and homework		4		4		4			4		4		4				24
3.	Quizzes					5							5					10
4.	Mid/end-terms								13							13		26
5.	Final examination																40	40
	Total	60																40 100

Grading policy:

1. Laboratory Works and Rules for the defense:

1. Components of the grade:

- Report Writing (50%):** The quality, clarity, and completeness of the lab report constitute 50% of the overall grade for the lab paper. Reports must follow the guidelines provided, including formatting and content structure.
- Oral Defense (50%):** The remaining 50% of the lab work grade will be based on the oral defense of the report. Students will be evaluated on their understanding of the complete lab work, their ability to explain their findings and answer questions during the defense.

2. Grading Criteria:

- Each component (report and oral defense) is graded on a scale of 0 to 2 points.
- The final grade for the laboratory work will be the sum of the points for both components, giving a maximum of 4 points. We will check your "Report write" works only after oral defense.

3. Deadline:

- Each lab paper must be completed and defended within two weeks of receiving the assignment.
- Students may defend their work before the next lab if they are willing to do so early.
- Important:** If students fail to turn in lab work before the next lab, the work will not be accepted and will be graded as 'not turned in'.

4. defense requirements:

- The defense of the laboratory work includes both presentation of the main material and answers to the instructor's questions.
- Students are recommended to prepare for the defense by thoroughly studying theoretical and practical aspects of the completed work.

2. Quizzes:

- Format:** Tests will consist of various types of questions including:
 - Questions with one correct answer.
 - Multiple-choice questions with multiple correct answers.

- Open-ended questions where students will be required to formulate their own answer.

These questions are designed to test understanding of key concepts covered in lectures and study materials.

- Grading Criteria:** The grade for each quiz will be determined by the number of correct answers. Each correct answer will contribute to the total quiz score.

3. Midterm/Endterm Exams:

- Format:** Both the midterm and endterm exams will include different types of questions:
 - Questions with one correct answer.
 - Multiple-choice questions with multiple correct answers.
 - Open-ended questions where students will have to formulate their own answer.

Exams will cover all material covered in the course up to the time of the exam.

- Grading Criteria:** The grade for the midterm and endterm exams will be determined by the number of correct answers. Like quizzes, each correct answer will count toward the overall exam grade. The final exam grade will be calculated as a percentage of the total number of correct answers.

4. Final Exam:

- Format:** Final exams will be conducted in a multiple-choice format, covering all material taught up to that point in the course.
- Grading Criteria:** The grade for the Final exams will be based on the number of correct answers. Like quizzes, each correct answer will count towards the total exam score. The final exam grade will be calculated as a percentage of the total correct answers.

Grade		Achievement percentage	Assessment criterion
«Excellent»	A	95-100%	This grade is given when the student: demonstrated a complete understanding of the course material; did not make any errors or inaccuracies; completed control and laboratory work in a timely and correct manner, and submitted reports on them; demonstrated original thinking; submitted control quizzes on time and without any errors; completed homework assignments; engaged in research work; independently used additional scientific literature in studying the discipline; was able to independently systematize the course material.
	A -	90-94%	
«Good»	B+	85-89%	This grade is given when the student: Has mastered the course material at no less than 75%; Did not make gross errors in responses; Timely completed control and laboratory work and submitted them without fundamental remarks;
	B	80-84%	

	B-	75-79%	Correctly completed and timely submitted control tests and homework assignments without fundamental remarks;
	C+	70-74%	Utilized additional literature as indicated by the instructor; Engaged in research work, made non-fundamental errors, and fundamental errors corrected by the student themselves; Managed to systematize the course material with the help of the instructor.
«Satisfactory»	C	65-69%	This grade is given when the student:
	C-	60-64%	Has mastered the course material no less than 50%;
	D+	55-59%	Required assistance from the instructor when completing control and laboratory work, homework assignments;
	D	50-54%	Made inaccuracies and non-fundamental errors when submitting control tests; Did not demonstrate activity in research work, relied solely on the educational literature indicated by the instructor; Experienced more difficulty in systematizing the material.

7. General Rules

Academic Integrity

- Academic integrity is of utmost importance in this course. Students are expected to adhere to the highest standards of honesty and integrity in their academic work.
- Any form of academic dishonesty, including plagiarism, cheating, unauthorized collaboration, and falsification of data or information, will not be tolerated.
- The use of AI tools for reports, assignments, projects, midterm/endpoint, quizzes and is restricted. Any work found to contain more than 15% AI-generated text will be automatically assessed as 0.
- Violations of academic integrity will result in serious consequences, including receiving a zero on the assignment, project, or exam in question, and may lead to further disciplinary actions as per the university's academic policies.
- Students are encouraged to seek clarification from the instructor if they are unsure about what constitutes academic dishonesty in any given context.

Attendance policy

- Regular attendance is crucial for success on this course. Active participation in lectures, labs, and discussions is expected from all students.
- Students must arrive on time for each session. Any late arrival will be recorded as an absence.
- Students are allowed a maximum of 30% absence throughout the course. This includes absences due to late arrivals.
- Exceeding the 30% absence limit will lead to failing the course.
- Students are responsible for catching up on any missed material or assignments due to absence. The instructor is not obliged to repeat lessons or provide special sessions for students who have missed classes.

- To complete the course, you must listen to a lecture of more than 80% on the platform "uni-x.kz" otherwise you will automatically fail it.

Office Hours:

1. **Purpose:** If a student cannot defend lab work during class time, he/she may do so during office hours.
2. **Advance Notice:** Students must notify the instructor of their intention to defend the paper by emailing the instructor in Microsoft Teams and providing a specific time and day of the week. This must be done at least one day prior to the intended meeting. The meeting will be confirmed only after the teacher responds and agrees on the time
3. **Meeting location and platform:**
 - a. Meetings will be held in room 268.
 - b. Pre-arrangement of time is done through Microsoft Teams. Notification must be sent as scheduled.
4. **Meeting Duration:** Each student will be given up to 10 minutes to defend the paper.
5. Students are expected to be punctual. Late arrivals will not be tolerated and missed time may affect the ability to defend the paper in the allotted time.

Instructor Nursultan

Nursultan Yergazizuly Alpysbay

Minutes # 1 of School of Information Technology and Engineering meeting on August 18, 2025